PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T36 Classified

Cross-session • chapter-organised candidate

13 classified items • 42 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T36 • 13 items • 42 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Graphing Inequalities

Unit 2 • 42 marks • 13 item(s)

June 2025 | 4MA1-2H Q9 | 3 marks

Derive the sloping boundary equation from the graph

June 2024 | 4MA1-2H Q2 | 4 marks

Boundaries drawn as an inseparable prerequisite

November 2023 | 4MA1-1H Q8 | 3 marks

Boundaries already supplied or easily inferred

June 2023 | 4MA1-1H Q7 | 4 marks

Derive the sloping boundary equation from the graph

November 2024 | 4MA1-2H Q7 | 4 marks

Boundaries drawn as an inseparable prerequisite

June 2022 | 4MA1-2H Q2 | 4 marks

Boundaries drawn as an inseparable prerequisite

January 2022 | 4MA1-2H Q13 | 3 marks

Derive the sloping boundary equation from the graph

January 2020 | 4MA1-1H Q10 | 3 marks

Derive the sloping boundary equation from the graph

January 2021 | 4MA1-2H Q4 | 3 marks

Boundaries drawn as an inseparable prerequisite

January 2021 | 4MA1-2H Q4 | 1 marks

Boundaries drawn as an inseparable prerequisite

November 2021 | 4MA1-2H Q13 | 3 marks

Derive the sloping boundary equation from the graph

January 2023 | 1H Q11 | 3 marks

Boundaries drawn as an inseparable prerequisite

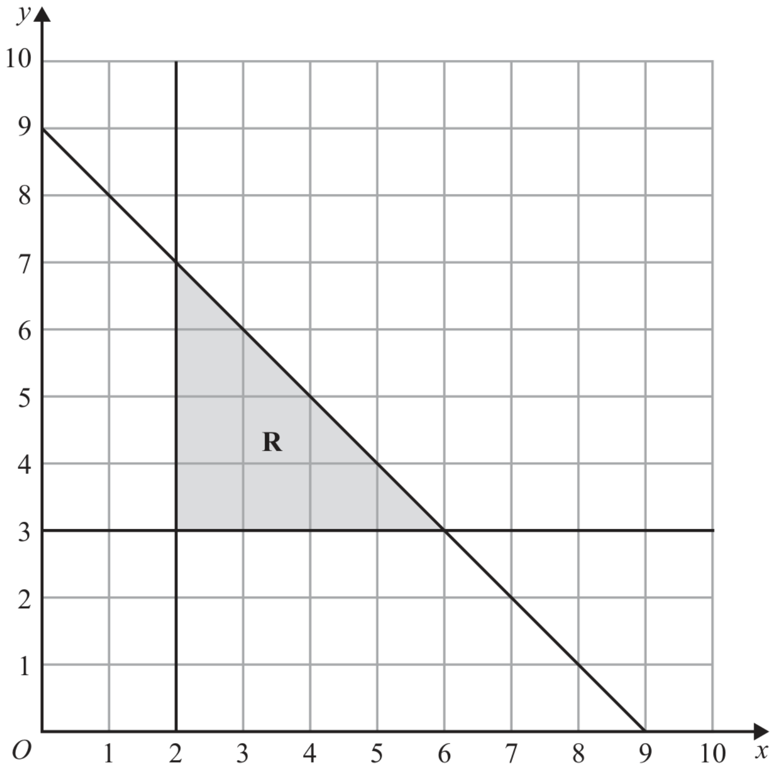
November 2025 | 2H Q10 | 4 marks

Boundaries drawn as an inseparable prerequisite

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down three inequalities that define the region **R**

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(3)

Worked Solution 25

Region of inequalities

UNIT 2
3 marks

Read the three boundaries

$$x = 2, \qquad y = 3, \qquad x + y = 9.$$

Choose the shaded side

$$R: \quad x \geq 2, \qquad y \geq 3, \qquad x + y \leq 9.$$

Final answer

$$x \geq 2, \qquad y \geq 3, \qquad x + y \leq 9$$

Question 13

4MA1-2H Q2

ORIGINAL QUESTION
4 marks

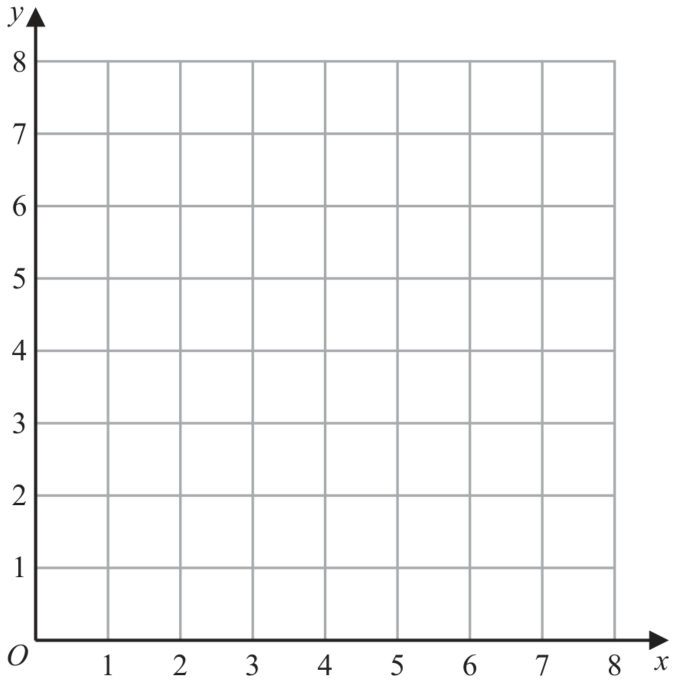
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geqslant 2 \qquad x \leqslant 6 \qquad y \leqslant x + 1$$

Label the region **R**

(1)

Worked Solution 13

Lines and a region of inequalities

MODULAR UNIT 2

4 marks

Boundaries

Draw and label $y = 2,\ x = 6,\ y = x + 1.$

Region

Shade the closed triangle with vertices $(1, 2), (6, 2), (6, 7)$ and label R.

Final answer

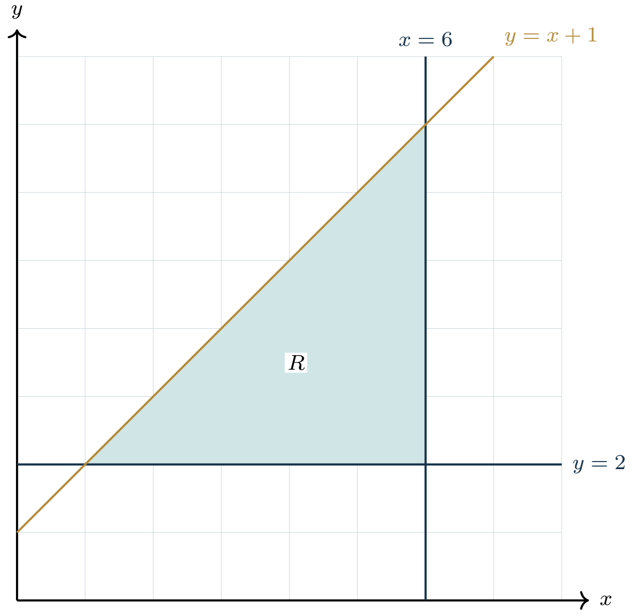
The triangular region R

Completed diagram 13

Lines and a region of inequalities

MODULAR UNIT 2
4 marks

Completed solution diagram: Lines and feasible region



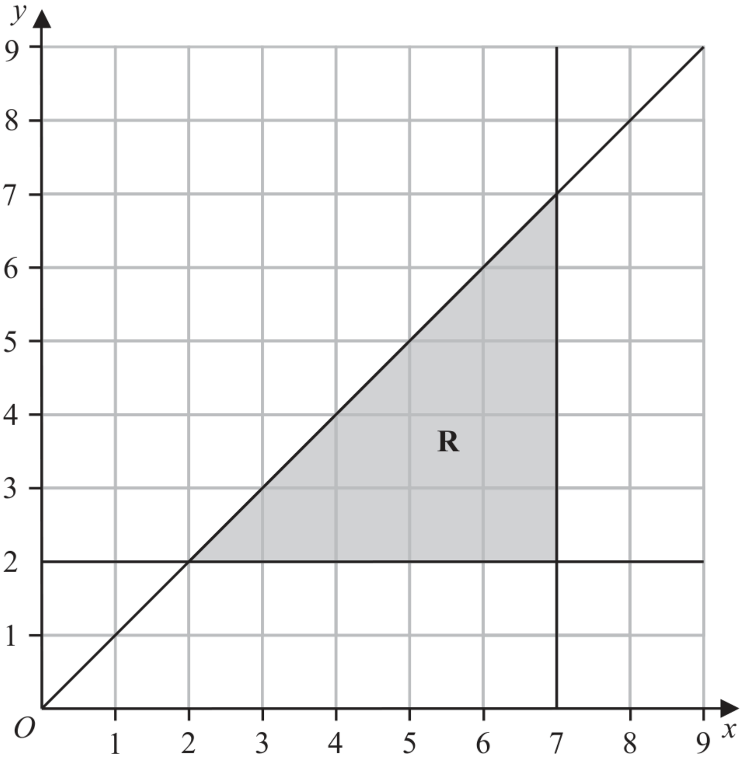
Question 05

4MA1-1H Q8 (b)

ORIGINAL QUESTION UNIT 2

3 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

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....

(3)

Worked solution 05

Define the shaded region

MODULAR UNIT 2

3 marks

(b) – Define the shaded region

Read each boundary

$$y = 2, \quad x = 7, \quad y = x.$$

Choose the shaded side

$$R: \ y \geq 2, \ x \leq 7, \ y \leq x.$$

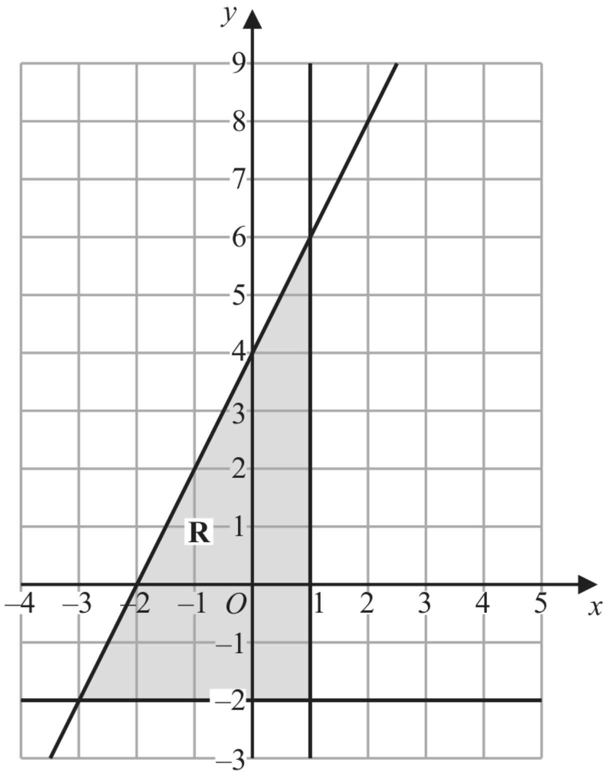
Final answer

$y \geq 2, x \leq 7, y \leq x$

Question 06

4MA1-1H Q7

ORIGINAL QUESTION
4 marks



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

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Worked Solution 06

Region inequalities

MODULAR UNIT 2

4 marks

Read the vertical boundary

$$x \leq 1.$$

Read the horizontal boundary

$$y \geq -2.$$

Find the sloping line and shaded side

$$y = 2x + 4, \qquad y \leq 2x + 4.$$

Final answer

$$x \leq 1, \quad y \geq -2, \quad y \leq 2x + 4$$

Original question 21

4MA1-2H Q7 M01, M02

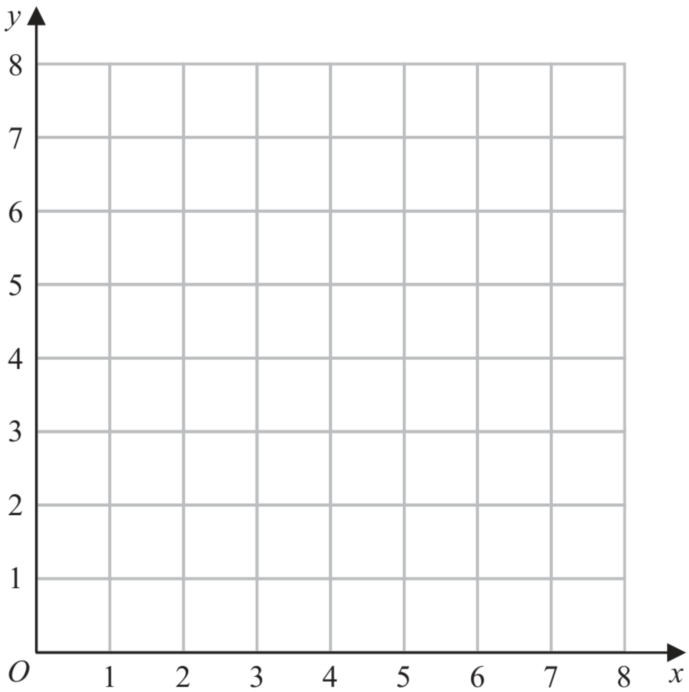
MODULAR UNIT 2

4 marks

7 (a) On the grid, draw the straight line with equation

(i) $x = 3$ (ii) $y = 1$ (iii) $x + y = 7$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$x \geqslant 3 \qquad y \geqslant 1 \qquad x + y \leqslant 7$$

Label the region **R**

(1)

Worked solution 21

Draw three straight lines / Shade the simultaneous-inequality region

MODULAR UNIT 2
4 marks

M01 – Draw three straight lines

Draw x=3
A vertical line through x=3.

Draw y=1
A horizontal line through y=1.

Draw x+y=7
A line of gradient -1 through the intercepts (0,7) and (7,0); label all three.

Final answer
The labelled lines x=3, y=1 and x+y=7

M02 – Shade the simultaneous-inequality region

Translate each inequality

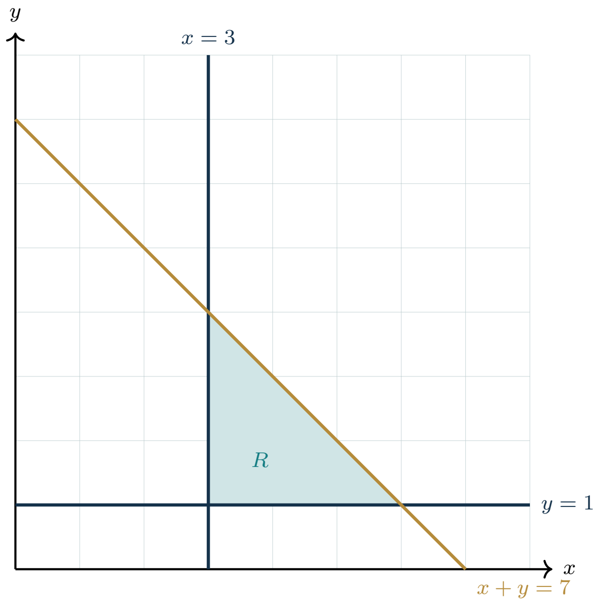
$$x \geq 3, \quad y \geq 1, \quad x + y \leq 7.$$

Shade and label
Shade the closed triangular region and label it R.

Final answer

$$R: \ x \geq 3, \ y \geq 1, \ x + y \leq 7$$

Completed solution diagram: Boundary lines and the Unit 2 feasible region


y
x = 3
R
y = 1
x
x + y = 7

Original question 21

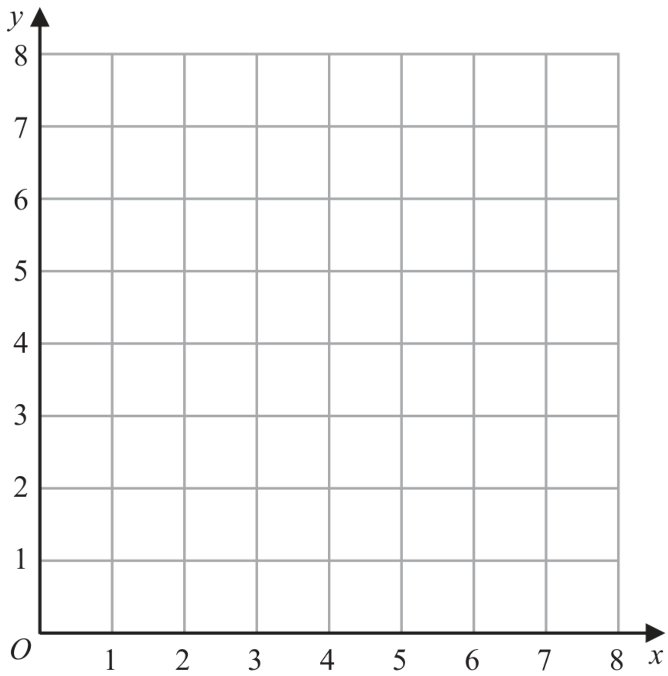
4MA1-2H Q2 Q2(a), Q2(b)

ORIGINAL QUESTION

4 marks

2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$ (ii) $x = 2$ (iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geqslant 1$ $x \geqslant 2$ $x + y \leqslant 7$

Label the region **R**.

(1)

Worked solution 21

Chapter: Linear Graphs ($y = mx + c$); Graphing Inequalities

MODULAR UNIT 2
4 marks

Q2(a) – Chapter: Linear Graphs ($y = mx + c$)

Family: Generate and plot a linear graph • Variant: Plot a line from generated values

Method
Draw and label the horizontal line $y = 1$.
Why: Every point on this line has y-coordinate 1.

Method
Draw and label the vertical line $x = 2$.
Why: Every point on this line has x-coordinate 2.

Method
Draw and label $x + y = 7$ through $(0, 7)$ and $(7, 0)$.
Why: Two intercepts determine the straight line.

Final answer
line 1: $y = 1$

Final answer
line 2: $x = 2$

Final answer
line 3: $x + y = 7$

Q2(b) – Chapter: Graphing Inequalities

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Method
Shade the region above $y = 1$, to the right of $x = 2$, and below $x + y = 7$; label it R.
Why: All three inequalities must hold simultaneously.

Final answer
region: R has vertices $(2, 1)$, $(2, 5)$, $(6, 1)$

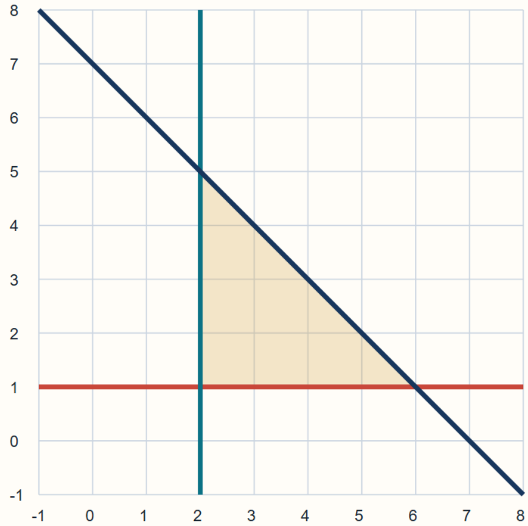
Completed diagram 21

Chapter: Linear Graphs (y = mx + c); Graphing Inequalities

MODULAR UNIT 2
4 marks

Q2 Completed Cartesian graph

Boundary lines for y=1, x=2 and x+y=7



$y = 1$

$x = 2$

$x + y = 7$

Region: $x \geq 2$, $y \geq 1$, $x + y \leq 7$

Vertices: (2,1), (6,1), (2,5)

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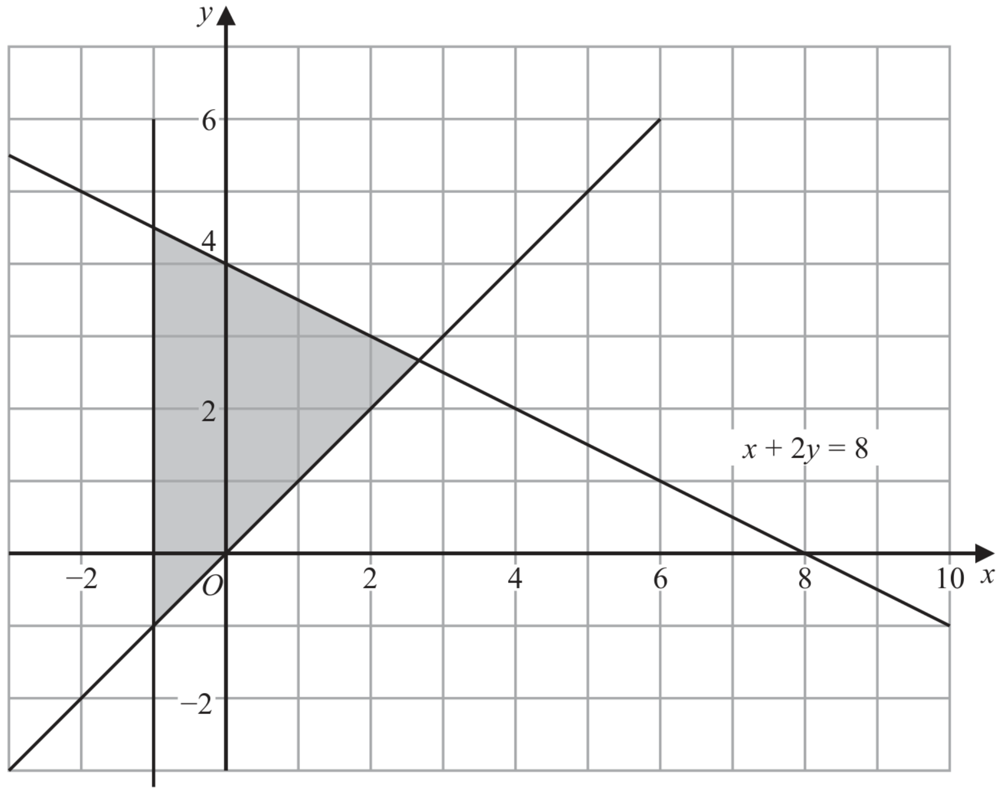
Original question 27

4MA1-2H Q13

ORIGINAL QUESTION
3 marks

13 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down three inequalities that define the shaded region.

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Worked solution 27

Chapter: Graphing Inequalities

MODULAR UNIT 2
3 marks

Q13 – Chapter: Graphing Inequalities

Family: Recover inequalities from a shaded region • Variant: Derive the sloping boundary equation from the graph

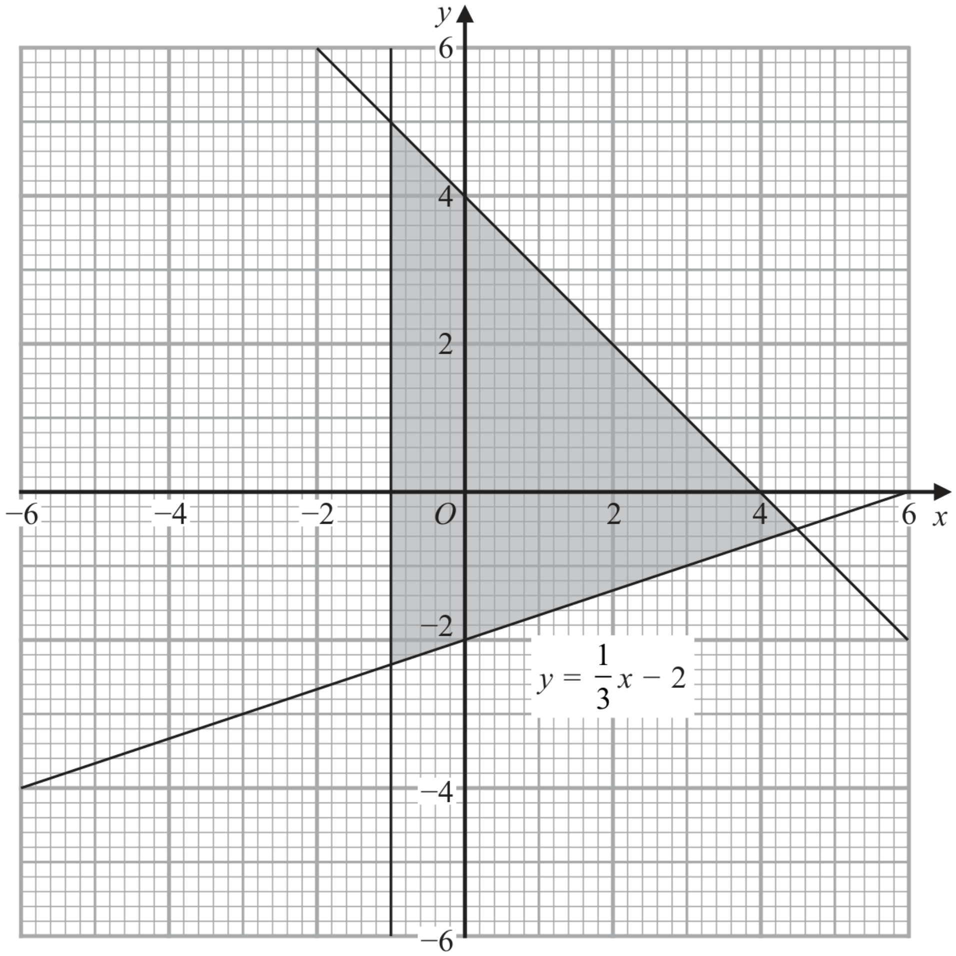
Method
The vertical boundary is $x=-1$ and the shaded side is $x\geq-1$.
Why: Test a point in the shaded region.

Method
The rising boundary is $y=x$ and the shaded side is $y\geq x$.
Why: Read the line and shaded side.

Method
The falling boundary is $x+2y=8$ and the shaded side is $x+2y\leq 8$.
Why: Use the labelled boundary equation and a test point.

Final answer
inequalities: $x\geq-1$, $y\geq x$, $x+2y\leq 8$

10 The shaded region in the diagram is bounded by three lines.
The equation of one of the lines is given.



Write down the three inequalities that define the shaded region.

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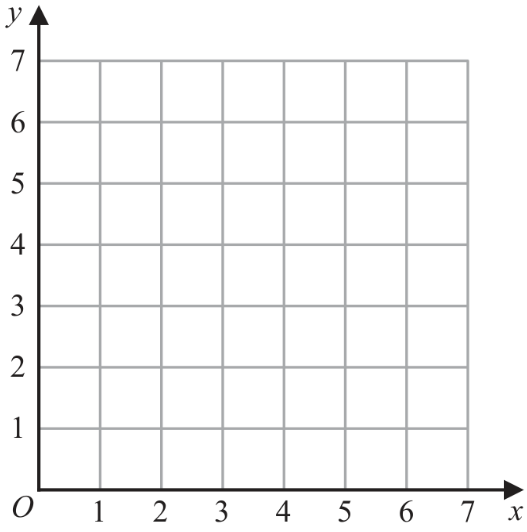
Worked solution

January 2020 | Question 10 | 3 marks

Family: Recover inequalities from a shaded region • Variant: Derive the sloping boundary equation from the graph

Method / working
1. The vertical boundary is x=-1 and the shaded side is to the right: x>=-1.
2. The oblique boundary x+y=4 has the shaded side below it: x+y<=4.
3. The given boundary y=(1/3)x-2 has the shaded side above it: y>=(1/3)x-2.
Final answer: x>=-1; x+y<=4; y>=(1/3)x-2

4



(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)

Worked solution

January 2021 | Question 4 | 3 marks

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Method / working

1. Draw the vertical line $x=1.5$.
2. Draw $y=x$ through the origin.
3. Draw $x+y=6$ through (0,6) and (6,0), labelling each line.

Final answer: Draw $x=1.5$, $y=x$ and $x+y=6$

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$$x \geqslant 1.5 \qquad y \geqslant x \qquad x + y \leqslant 6$$

Label the region **R**.

(1)

Worked solution

January 2021 | Question 4 | 1 marks

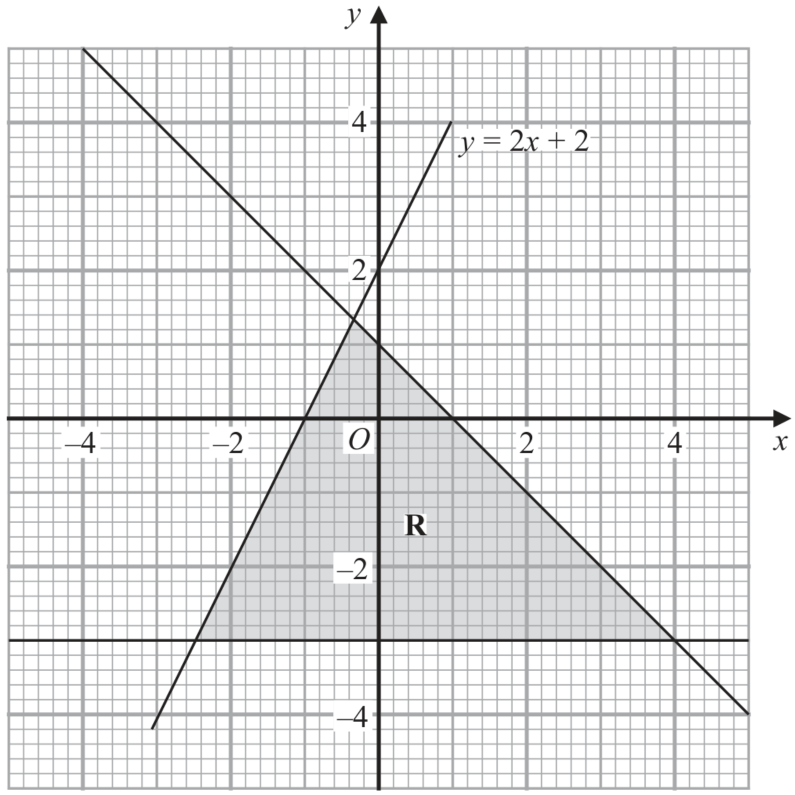
Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Method / working

1. Shade the intersection of the three half-planes and label it R.

Final answer: Region R: x>=1.5, y>=x, x+y<=6

13



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Write down the three inequalities that define **R**.

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Worked solution

November 2021 | Question 13 | 3 marks

Family: Recover inequalities from a shaded region • Variant: Derive the sloping boundary equation from the graph

Method / working

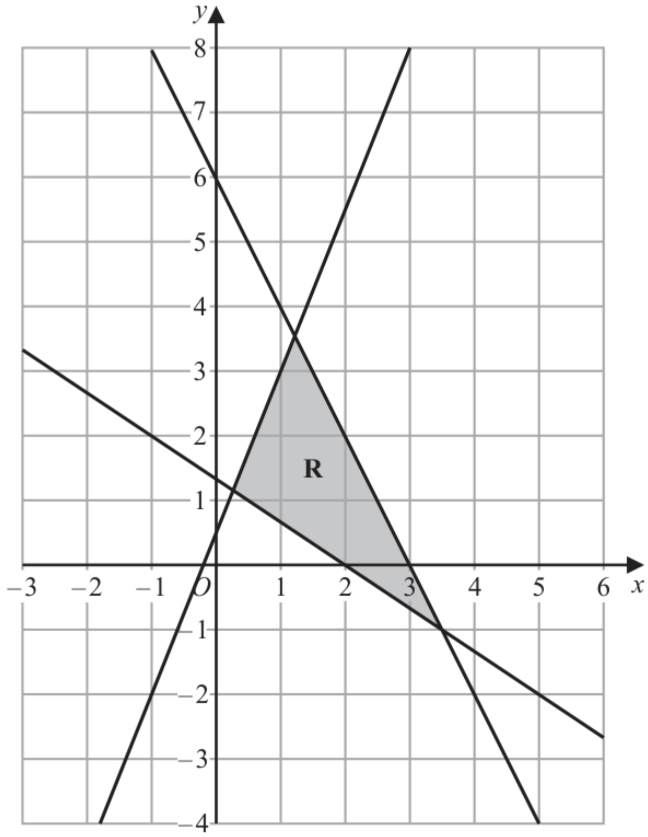
1. The bottom boundary is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. $y=-3$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. so
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. $y\geq-3$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. The rising boundary is
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. $y=2x+2$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. so the region has
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. $y\leq2x+2$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. The falling boundary goes through $(0,1)$ and $(1,0)$, so
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. $x+y=1$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. The region is below this line:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. $x+y\leq1$
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. Answer: $y\geq-3$, $y\leq2x+2$, $x+y\leq1$.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $y\geq-3$, $y\leq2x+2$, $x+y\leq1$.

January 2023 | 1H | Question 11 | 3 marks

Chapter OL-T36: Graphing Inequalities

11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

$2x + y = 6$

$2y = 5x + 1$

$3y + 2x = 4$

Write down the three inequalities that define **R**

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(Total for Question 11 is 3 marks)

Answer reference | January 2023 | Question 11

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

11		$2x+y\le 6$ $2y\le 5x+1$ $3y+2x\ge 4$	3	B3 oe for all three correct (B2 oe for any two correct) (B1 oe for any one correct) $2x+y\le 6$ equivalent to $y\le -2x+6$ oe $2y\le 5x+1$ equivalent to $y\le 2.5x+0.5$ oe $3y+2x\ge 4$ equivalent to $y\ge -\frac{2}{3}x+\frac{4}{3}$ oe Allow the following inequalities $2x+y<6$ oe $2y<5x+1$ oe $3y+2x>4$ oe
				Total 3 marks

Worked solution

January 2023 | Question 11 | 3 marks

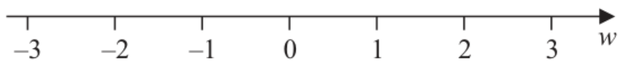
Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Official final mark-scheme pages are embedded immediately after this question.

November 2025 | 2H | Question 10 | 4 marks

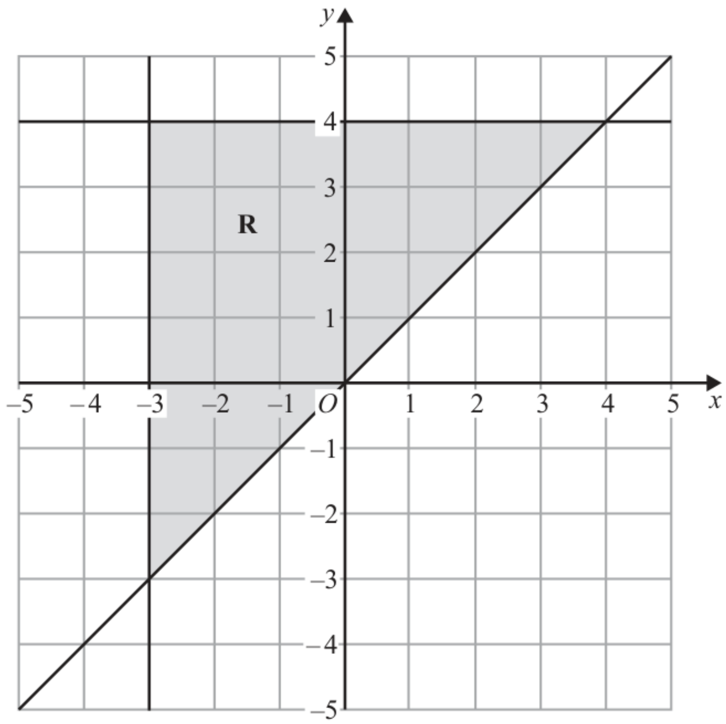
Chapter OL-T36: Graphing Inequalities

10 (a) On the number line, represent the inequality $w < 1$



(1)

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

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(3)

(Total for Question 10 is 4 marks)

Worked solution

November 2025 | Question 10 | 4 marks

Family: Construct the feasible region from inequalities • Variant: Boundaries drawn as an inseparable prerequisite

Solution
For part (a), w<1 is shown with an open circle at 1 and an arrow to the left.
For part (b), the region is right of x=-3, below y=4, and above y=x:
x>=-3
y<=4
y>= x
Answer: (a) Open circle at 1, shade left. (b) x>=-3, y<=4, y>= x.

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION